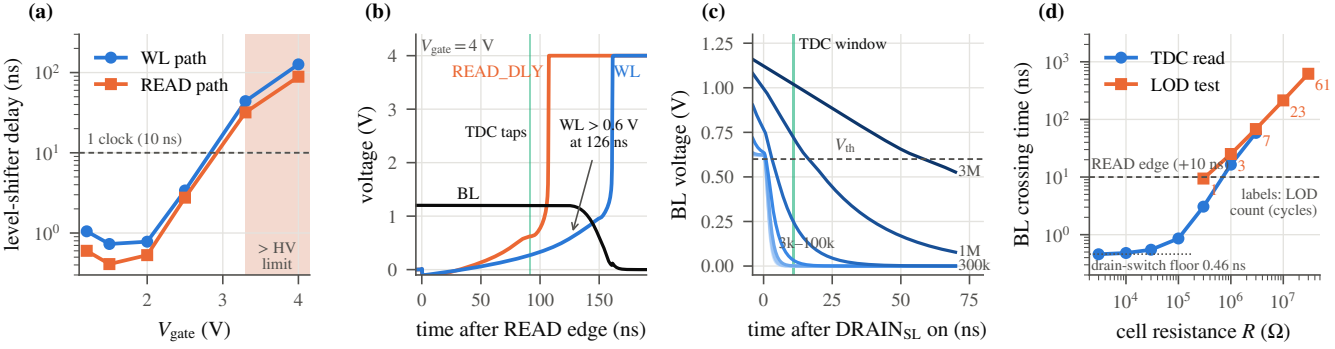
(a)
WL path
READ path
1 clock (10 ns)
> HV limit
level-shifter delay (ns)
V_{gate} (V)
(b)
$V_{gate} = 4$ V
READ_DLY
WL
TDC taps
WL > 0.6 V at 126 ns
BL
voltage (V)
time after READ edge (ns)
(c)
TDC window
V_{th}
3M
1M
300k
8k–100k
BL voltage (V)
time after DRAIN$_{SL}$ on (ns)
(d)
TDC read
LOD test
READ edge (+10 ns)
drain-switch floor 0.46 ns
labels: LOD count (cycles)
61
23
7
3
BL crossing time (ns)
cell resistance R (Ω)